

ORIGINAL ARTICLE

Social deprivation and 1-year survival after stroke: a prospective cohort study

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Abstract

Background: Social deprivation may have a deleterious influence on post-stroke outcomes, but available data in the literature are mixed.

Aim: The aim of this cohort study was to evaluate the impact of social deprivation on 1-year survival in patients with first-ever stroke.

Methods: Social deprivation was assessed at individual level with the EPICES score, a validated multidimensional questionnaire, in 1312 patients with ischemic stroke and 228 patients with spontaneous intracerebral hemorrhage, who were prospectively enrolled in six French study centers. Baseline characteristics including stroke severity and pre-stroke functional status were collected. Multivariable Cox models were generated to evaluate the associations between social deprivation and survival at 12 months in ischemic stroke and intracerebral hemorrhage separately.

Results: A total of 819 patients (53.2%) were socially deprived (EPICES score ≥ 30.17). In ischemic stroke, mortality at 12 months was higher in deprived than in non-deprived patients (16% vs. 11%, $p = 0.006$). In multivariable analyses, there was no association between deprivation and death occurring within the first 90 days following ischemic stroke (adjusted hazard ratio [aHR] 0.81, 95% CI 0.54–1.22, $p = 0.32$). In contrast, an excess in

mortality was observed between 90 days and 12 months in deprived compared with non-deprived patients (aHR 1.97, 95% CI 1.14–3.42, $p = 0.016$). In patients with intracerebral hemorrhage, mortality at 12 months did not significantly differ according to deprivation status.

Conclusions: Social deprivation was associated with delayed mortality in ischemic stroke patients only and, although the exact underlying mechanisms are still to be identified, our findings suggest that deprived patients in particular may benefit from an optimization of post-stroke care.

KEYWORDS

infarction, intracerebral hemorrhage, cohort studies, prognosis, psychosocial deprivation

INTRODUCTION

The impact of social deprivation on post-stroke outcomes is mixed. Although several studies conducted in different countries have suggested that neighborhood deprivation could be associated with increased mortality after stroke [1–3], other studies have concluded that the association is only marginal [4] or not significant [5,6]. In addition, varying aspects of social deprivation have been investigated. Hence, an inverse association between income and post-stroke mortality has been reported in the literature [7–10], but these associations were inconsistently found in women [11–13] or not demonstrated in some populations [14]. Similarly, lower education levels have been associated with a higher risk of death [7,8], but only in some age categories for women [12], and other studies did not confirm this result [10,14]. Finally, unemployment status and low-income socioprofessional categories have been associated with higher mortality in several studies [8,10,14].

There are multiple potential reasons for these inconsistent findings. Among these, a major limitation of previous studies is that social deprivation was measured either at a population-level based on aggregated data, or using individual but limited indicators. Moreover, most studies did not take into account important confounding variables in the analyses. For instance, stroke severity at onset was not included even though it has been strongly associated with deprivation [15]. Finally, some studies did not distinguish between ischemic and hemorrhagic strokes despite their very different prognoses.

Therefore, the aim of this study was to evaluate the impact of social deprivation assessed with a multidimensional score on 1-year survival in patients with first-ever stroke.

METHODS

Study population and setting

The INDIA (INégalités sociales et pronostic des accidents vasculaires cérébraux à Dijon et en Antilles-Guyane) study is a French multicenter prospective cohort study including patients aged > 18 years admitted for acute stroke (ischemic stroke, spontaneous intracerebral hemorrhage or subarachnoid hemorrhage)

in any one of the six participating French centers: one in mainland France (Dijon, Burgundy) and five in French Overseas Departments (Fort-de-France, Martinique; Pointe-à-Pitre, Guadeloupe; Cayenne, Saint Laurent du Maroni, and Kourou, French Guiana). Patients were eligible if they could be interviewed either personally or, in the case of aphasia or cognitive or consciousness disorders, via a next of kin. Patients with a history of symptomatic stroke or other short-term life-threatening diseases were not included, as well as patients (or next of kin) who were planned not to be reachable by telephone during the follow-up period. For the present analysis, patients with subarachnoid hemorrhage ($n = 14$) were further excluded. Details of the design and methods of the INDIA study have been described previously [15].

By varying the deprivation rate from 30% to 70%, with an average rate of death of 12% at 1 year in non-deprived patients, 1454 analyzable patients were required to detect a minimal increase of 6 points (i.e. a hazard ratio [HR] of 1.55) with a power of 80% and an alpha risk of 0.05 (two-sided test).

The study protocol was approved by the Burgundy Ethics Committee (CPP Est 1, 16 May 2010) and the National Commission for Data Processing and Civil Liberties (15 April 2011). According to French law relative to observational studies, all patients received written information about the study, but signed consent was not required. The study was registered with ClinicalTrials.gov under the identifier NCT01906463.

Assessment of social deprivation

The main variable of interest was deprivation, which was assessed at the individual level using the EPICES (Evaluation de la Précarité et des Inégalités de santé dans les Centres d'Examen de santé) score validated on a French cohort of 197,389 subjects [16]. This score measured at baseline and reflecting the social and material dimensions of deprivation has been shown to be a reliable tool to measure individual deprivation and to be strongly correlated with the Townsend and Carstairs indices [17]. The EPICES score is composed of 11 binary questions with a focus on marital status, health insurance status, economic status, family/social support and leisure activity during the last 12 months (Table 1). Each item is associated with

TABLE 1 Items of the EPICES^a score

Question	Coefficient ^b
1. Do you sometimes meet a social worker?	10.06
2. Do you have complementary health insurance? ^c	-11.83
3. Do you live as a couple?	-8.28
4. Are you a homeowner?	-8.28
5. Are there periods in the month when you have real financial difficulties to meet your needs (food, rent, electricity)?	14.80
6. Have you done any sports activities in the last 12 months?	-6.51
7. Have you been to any shows in the last 12 months?	-7.10
8. Have you been on holiday in the last 12 months?	-7.10
9. Have you seen any family member in the last 6 months (other than your parents or children)?	-9.47
10. If you have difficulties, is there anyone around you who could take you in for a few days?	-9.47
11. If you have difficulties, is there anyone around you who could provide you with material assistance?	-7.10
Intercept	75.14

^aEPICES: Evaluation de la Précarité et des Inégalités de santé dans les Centres d'Examen de Santé.

^bCoefficient used to calculate the EPICES score: each question coefficient is added to intercept if the answer is 'yes'. The score varies from 0 (no deprived) to 100 (most deprived). Questions were translated from French to English.

^cIn France, complementary private insurance is needed to complete the basic health insurance that only partially reimburses medical expenses. Persons whose income is below a resource threshold can apply for public complementary universal medical cover.

a weight determined by principal component analysis. This score ranges from 0 to 100, from the lowest to the highest level of social deprivation, but it is mostly used as a dichotomous variable to discriminate deprived and non-deprived patients, with patients with a score ≥ 30.17 considered as deprived [16].

Data collection, follow-up and outcome assessment

Specially trained clinical research technicians collected all data prospectively, with the assistance of the doctors responsible for patient management. Sociodemographic characteristics, preexisting conditions, clinical data at baseline, and acute stroke therapy were collected. Stroke severity at admission was assessed using the National Institutes of Health Stroke Scale (NIHSS) score. The degree of disability/dependence before stroke was measured with the premorbid modified Rankin Scale (mRS).

Follow-up phone calls (or visits if the patient was still hospitalized) took place 1, 6 and 12 months after stroke. Data regarding location (hospital, long-stay residential institution, rehabilitation center or home), stroke recurrence and re-hospitalization were collected.

Survival was determined by follow-up, including contacting a next of kin or the patient's physician if the patient could not be reached. For patients lost to follow-up, the National Identification Register of Individuals was contacted.

Statistical analysis

Analyses were conducted separately for ischemic stroke and spontaneous intracerebral hemorrhage, seeing as prognosis differs considerably according to stroke type. A complete case analysis was done when the rate of missing data was lower than 5%. Otherwise, a category 'missing data' was generated for the concerned covariate. Patient characteristics (according to deprivation status) were compared using a chi-squared test or Fisher's test for qualitative variables, and using a t-test or Wilcoxon Mann-Whitney test for quantitative variables, as appropriate. The primary outcome was survival at 12 months, defined as the time from stroke until death from any cause. Survival between groups was plotted using the Kaplan-Meier method and compared using the log-rank test stratified on the center. Candidate variables for multivariable analysis were based on clinical relevance, and included age, sex, body mass index (BMI), ethnic origin, home ownership, hypertension, diabetes, hypercholesterolemia, chronic heart failure, alcohol, smoking, atrial fibrillation, prior transient ischemic attack, prior myocardial infarction, pre-stroke treatment (antiplatelet therapy, anticoagulant therapy, antihypertensive drugs), Trial of Org 10172 in Acute Stroke Treatment stroke classification, admission NIHSS score and thrombolysis. Variables associated with survival at 12 months with a p value < 0.20 in the univariate analysis (log-rank tests) were included as covariates in a multivariable Cox model with a stepwise selection procedure (p value < 0.20 for entry, p value < 0.20 to stay in the model). The assumption of proportionality was assessed using the supremum test for proportional hazards assumption [18]. In case of violation of this assumption, a piecewise constant HR was calculated over distinct periods in which a proportional hazards assumption was met [19]. The variables retained in the final models were age, NIHSS, atrial fibrillation, diabetes, BMI and chronic heart failure. Posterior adjustments were performed on pre-stroke mRS and high school diploma to assess the EPICES effect modification. Sensitivity analyses were performed with further adjustment for sex and intravenous (IV) thrombolysis (not retained by the selection procedure) due to their clinical relevance, and completed by considering EPICES score as a continuous variable. All analyses were stratified on the center for accounting potential center effects. The significance level was set at $p < 0.05$. Analyses were performed using SAS 9.4.

RESULTS

Between June 2011 and October 2014, 1573 consecutive patients were included in the INDIA study with a median follow-up time of 373 days (IQR 366–414). Patients with subarachnoid hemorrhage

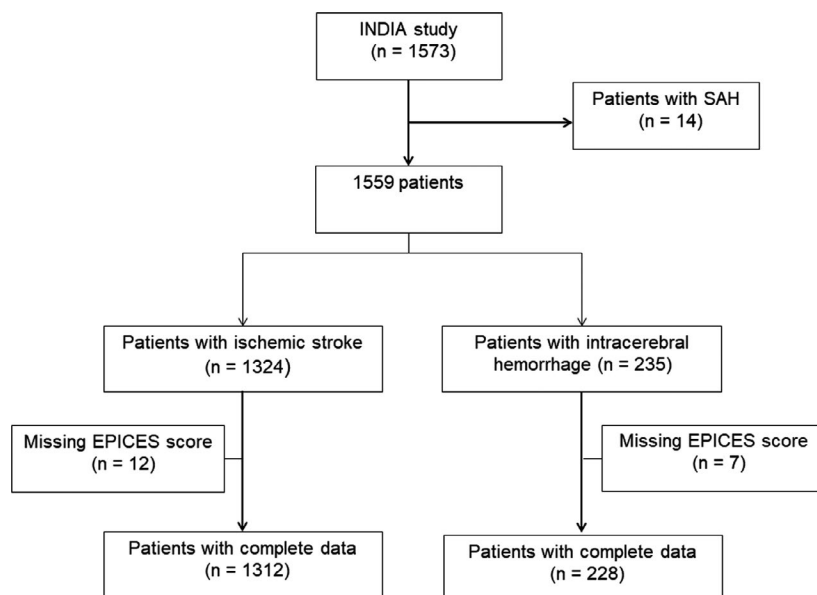


FIGURE 1 Flow chart of INDIA study. EPICES, Evaluation de la Précarité et des Inégalités de santé dans les Centres d'Examen de Santé; SAH, subarachnoid hemorrhage

($n = 14$) were excluded from the present analysis (Figure 1). Information on deprivation was available for 1540 (99%) patients.

Patients with ischemic stroke

A total of 1312 patients with ischemic stroke and available data about social deprivation were recruited (mean age 68 ± 15 years, 56% men). Slightly more than half of the patients with ischemic stroke were socially deprived (52%). The characteristics of deprived and non-deprived patients are presented in Table 2. Only 26% of deprived patients received IV thrombolysis, compared with 34% of non-deprived patients ($p = 0.001$), with a higher median time from stroke to admission for deprived patients than for non-deprived patients (3.4 h [IQR 1.7–9.8] vs. 2.6 h [1.4–6.7], respectively, $p = 0.0006$). When this analysis was restricted to patients admitted within 4.5 h after stroke (data available for 1056 patients), this difference was no longer significant (IV thrombolysis administered in 46% of deprived patients and 52% of non-deprived patients, $p = 0.14$). The median hospital stay differed between deprived and non-deprived patients (11 days [IQR 7–17] vs. 9 days [IQR 5–14], $p < 0.0001$). Fifty-eight patients (4%) died during the acute hospital stay, with no difference between groups ($p = 0.89$). Among the patients alive at discharge, 46% went back home, with no significant difference between deprived and non-deprived patients (45% vs. 48% respectively, $p = 0.23$).

During follow-up, non-deprived patients had higher post-stroke survival (Figure 2, $p = 0.0056$). The cumulative probability of death at 1 year was 16% in deprived patients vs. 11% in non-deprived patients (Table 3). Other variables associated with survival and retained as candidate covariates in the multivariable analysis ($p < 0.20$) are presented in the Table S1. The assumption of proportional hazards for the Cox model was met for all covariates except for deprivation

($p = 0.033$). Visual inspection of survival curves and score process trajectories revealed a cut-off at 90 days after stroke. The proportional hazards assumption was not violated within each of the two distinct periods (before and after 90 days). As a consequence, two distinct HRs were calculated before and after 90 days. The risk of death did not significantly differ according to deprivation status during the 90 days following stroke (crude HR 1.18 [95% CI 0.80–1.73], $p = 0.40$) but socioeconomic deprivation had an impact on the risk of death after 90 days (Table 4). The crude HR showed a 2.6-fold increase in the risk of death (HR 2.56, 95% CI 1.47–4.45, $p = 0.0009$) in deprived patients after 90 days compared with non-deprived patients. After adjustment for potential determinants of stroke prognosis (age, severity at admission and comorbidities), risk of death was nearly twice as high in deprived than in non-deprived patients (Model 1: HR 1.97 [95% CI 1.14–3.42], $p = 0.016$). Deprivation remained independently associated with the risk of death after further adjustment for pre-stroke mRS, although the magnitude was reduced (Model 2: HR 1.77, 95% CI 1.01–3.09, $p = 0.044$) and after adjustment for high school diploma (Model 3: HR 2.00, 95% CI 1.13–3.52, $p = 0.016$). Sensitivity analyses led to consistent results after including thrombolysis and sex in the analysis (Table S2), and when EPICES score was considered as a continuous variable (aHR 0.99 [95% CI 0.98–1.01], $p = 0.081$ before 90 days and aHR 1.02 [95% CI 1.01–1.03], $p = 0.015$ after 90 days).

Patients with intracerebral hemorrhage

A total of 228 patients with spontaneous intracerebral hemorrhage and available data about deprivation were included. Among them, 59% of patients were deprived. The characteristics of deprived and non-deprived patients are presented Table S3. The median hospital length of stay was shorter for deprived patients than for

Characteristic	Non-deprived patients (EPICES < 30.17) (N = 628)	Deprived patients (EPICES ≥ 30.17) (N = 684)	p value
Age (years)			
<50	88 (14%)	100 (15%)	0.003
50–65	160 (25%)	189 (28%)	
65–80	257 (41%)	218 (32%)	
≥80	123 (20%)	177 (26%)	
Male gender	350 (56%)	385 (56%)	0.84
BMI (kg/m ²)			
<20	24 (4%)	36 (5%)	<.001
20–25	196 (31%)	189 (28%)	
25–30	239 (38%)	200 (29%)	
≥30	99 (16%)	147 (21%)	
Unknown	70 (11%)	112 (16%)	
Hypertension	382 (61%)	447 (65%)	0.08
Diabetes	136 (22%)	188 (28%)	0.014
Hypercholesterolemia	167 (27%)	144 (21%)	0.018
Chronic heart failure	32 (5%)	63 (9%)	0.004
Alcohol ≥ 2 drinks/day	78 (12%)	83 (12%)	0.91
Current smoker	80 (13%)	125 (18%)	0.005
Atrial fibrillation	111 (18%)	118 (17%)	0.85
Prior transient ischemic attack	36 (6%)	38 (6%)	0.88
Prior myocardial infarction	27 (4%)	23 (3%)	0.37
Pre-stroke mRS score			
0	558 (89%)	540 (79%)	<.001
1–2	45 (7%)	54 (8%)	
3–4	25 (4%)	88 (13%)	
Ischemic stroke classification			
Lacunar stroke	59 (9%)	84 (12%)	0.045
Large-artery atheroma	97 (15%)	102 (15%)	
Cardiac embolism	173 (28%)	168 (25%)	
Other determined cause	31 (5%)	17 (2%)	
Undetermined cause	268 (43%)	313 (46%)	
Admission NIHSS score ^a			
0–3	241 (39%)	223 (33%)	0.08
4–9	181 (29%)	224 (33%)	
≥10	199 (32%)	227 (34%)	
IV thrombolysis tPA	213 (34%)	176 (26%)	0.001

Abbreviations: BMI, body mass index; EPICES, Evaluation de la Précarité et des Inégalités de santé dans les Centres d'Examen de Santé; IV, intravenous; mRS, modified Rankin Scale score; NIHSS, National Institutes of Health Stroke Scale; tPA, tissue-type plasminogen activator.

^aData were missing for 17 patients.

non-deprived patients (9 days [IQR 6–15] vs. 12 days [IQR 8–24], $p = 0.009$). In-hospital mortality rates were similar (11% of deprived patients vs. 14% of non-deprived patients, $p = 0.52$). In-hospital medical management did not significantly differ according to deprivation

status. The cumulative probability of death at 12 months was 20.4%, without significant difference between deprivation status (Table 3), and the survival curves did not differ significantly either (Figure S1, $p = 0.40$).

TABLE 2 Baseline characteristics of patients with ischemic stroke according to deprivation status

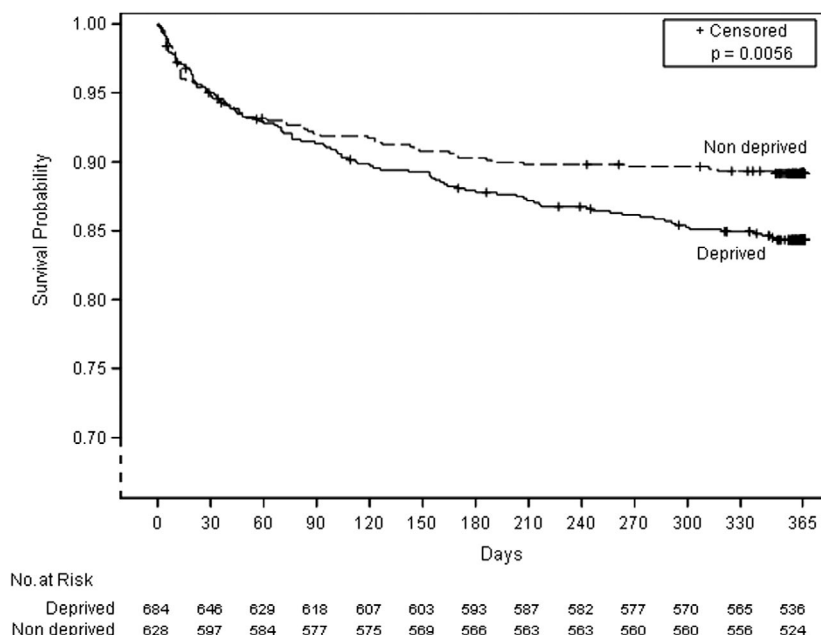


FIGURE 2 Survival after ischemic stroke according to deprivation status

TABLE 3 Cumulative probabilities of death at specified time points

Stroke type	At 1 month		At 3 months		At 6 months		At 12 months	
	%	95% CI	%	95% CI	%	95% CI	%	95% CI
Ischemic stroke								
Non-deprived patients (EPICES score < 30.17)	5.25	3.51–6.99	7.96	5.84–10.08	9.72	7.41–12.03	10.84	8.41–13.27
Deprived patients (EPICES score ≥ 30.17)	4.98	3.35–6.61	8.67	6.55–10.79	12.22	9.75–14.69	15.66	12.92–18.4
All	5.11	3.92–6.3	8.33	6.83–9.83	11.02	9.32–12.72	13.34	11.5–15.18
Intracerebral hemorrhage								
Non-deprived patients (EPICES score < 30.17)	16.13	8.66–23.6	18.28	10.42–26.14	20.43	12.24–28.62	22.6	14.09–31.11
Deprived patients (EPICES score ≥ 30.17)	8.89	4.09–13.69	14.96	8.9–21.02	16.48	10.19–22.77	18.88	12.2–25.56
All	11.85	7.66–16.04	16.3	11.5–21.1	18.08	13.06–23.1	20.38	15.13–25.63

Abbreviation: CI, confidence interval.

DISCUSSION

Our findings indicate that social deprivation is associated with reduced survival in patients with ischemic stroke. However, this association depended on the time after stroke. We observed an excess in mortality for deprived patients after 90 days following stroke (delayed mortality), but there was no difference in mortality within the first 90 days (early mortality) between deprived and non-deprived patients. In patients with intracerebral hemorrhage, deprivation status was not associated with 12-month survival.

Several explanations could account for the distinct impact of social deprivation on early and delayed mortality in patients with ischemic stroke. Although certain of the baseline characteristics for

deprived and non-deprived patients differed, these factors were taken into consideration in our multivariable models, and the adjustments did not alter our results. IV thrombolysis with tissue-type plasminogen activator (tPA) was less frequently performed in deprived patients than in non-deprived patients, but this difference could be explained by a later delay between stroke and admission for non-deprived patients. Moreover, IV thrombolysis with tPA was not associated with risk of death in our study, and the impact of deprivation in delayed mortality persisted after adjustment for thrombolysis. The absence of difference in early mortality suggests that acute management of patients was similar whatever their deprivation status. This was probably because our patients were recruited from public hospitals, and the French health system offers universal coverage for all citizens and legal residents, regardless of age or economic situation.

TABLE 4 Factors associated with death within 12 months following ischemic stroke

Factor	Model 1		Model 2		Model 3	
	aHR (95% CI)	p value	aHR (95% CI)	p value	aHR (95% CI)	p value
Deprivation		0.0456		0.054		0.0502
EPICES score < 30.17	1		1		1	
EPICES score ≥ 30.17						
Between 0 and 90 days	0.90 (0.61–1.34)	0.617	0.78 (0.52–1.17)	0.232	0.92 (0.62–1.38)	0.692
After 90 days	1.97 (1.14–3.42)	0.016	1.77 (1.01–3.09)	0.044	2.00 (1.13–3.52)	0.016
Age (years)		<0.0001		<0.0001		<0.0001
<50	1		1		1	
50–65	1.84 (0.62–5.48)	0.271	1.76 (0.59–5.23)	0.310	1.78 (0.60–5.30)	0.298
65–80	4.15 (1.49–11.58)	0.006	3.51 (1.26–9.84)	0.017	3.75 (1.34–10.52)	0.012
≥80	8.82 (3.17–24.53)	<0.0001	6.19 (2.19–17.46)	0.0006	7.96 (2.84–22.27)	<0.0001
Admission NIHSS score		<0.0001		<0.0001		<0.0001
0–3	1		1		1	
4–9	2.58 (1.46–4.56)	0.001	2.49 (1.40–4.41)	0.002	2.51 (1.42–4.46)	0.0016
≥10	6.68 (4.00–11.16)	<0.001	6.26 (3.75–10.45)	<0.0001	6.48 (3.87–10.86)	<0.0001
Atrial fibrillation	1.62 (1.13–2.32)	0.009	1.61 (1.12–2.32)	0.011	1.57 (1.09–2.26)	0.0157
Diabetes	1.55 (1.11–2.17)	0.011	1.56 (1.11–2.19)	0.009	1.57 (1.11–2.21)	0.010
BMI (kg/m ²)		0.073		0.085		0.167
<20	1.45 (0.79–2.64)	0.227	1.49 (0.81–2.71)	0.196	1.23 [0.65–2.33]	0.515
20–25	1		1		1	
25–30	0.88 (0.57–1.35)	0.553	0.90 (0.58–1.39)	0.631	0.83 (0.54–1.30)	0.424
≥30	1.04 (0.64–1.70)	0.868	0.94 (0.57–1.54)	0.800	1.06 (0.65–1.73)	0.821
Missing	1.66 (1.04–2.64)	0.034	1.60 (1.00–2.56)	0.049	1.53 (0.95–2.46)	0.080
Chronic heart failure	1.84 (1.20–2.81)	0.0049	1.73 (1.13–2.65)	0.011	1.91 (1.24–2.96)	0.004
Pre-stroke mRS				<0.0001		
0			1			
1–2			2.34 (1.53–3.59)	<0.0001		
3–4			2.19 (1.43–3.35)	0.0003		
High school diploma					0.60 (0.33–1.10)	0.101

Note: Model 1: adjusted for age, NIHSS score, atrial fibrillation, diabetes, BMI and chronic heart failure, and stratified on center.

Model 2: adjusted for age, NIHSS score, atrial fibrillation, diabetes, BMI, chronic heart failure and pre-stroke mRS score, and stratified on center.

Model 3: adjusted for age, NIHSS score, atrial fibrillation, diabetes, BMI, chronic heart failure and high school diploma, and stratified on center.

Abbreviations: aHR, adjusted hazard ratio; CI, confidence interval; EPICES, Evaluation de la Précarité et des Inégalités de santé dans les Centres d'Examen de Santé; BMI, body mass index; mRS, modified Rankin Scale score; NIHSS, National Institutes of Health Stroke Scale.

The rates of in-hospital mortality and of patients admitted to post-stroke rehabilitation facilities were very similar for deprived and non-deprived patients, reinforcing the hypothesis that patients in French hospitals are treated equally, whatever their socioeconomic status.

We showed in a previous population-based study that the impact of increasing neighborhood deprivation on post-stroke mortality was noticeable only after acute hospital discharge [1]. However, this previous study was limited by the fact that deprivation was evaluated using aggregated data. The present findings are consistent with our previous work, which raises questions about the reasons underlying the observed effects. Although

many variables were collected, we cannot exclude the possibility that some unmeasured contributors of delayed mortality, such as poorer general health or medication compliance, may have played a role. In addition, the quality of longer-term management of deprived patients, which may include suboptimal control of vascular risk factors or lower adherence to treatments [20], could be potential determinants.

In deprived patients a lack of social support, which is one of the domains of the EPICES score, may also have contributed to the excess in delayed mortality after ischemic stroke. Such adverse contextual conditions have been shown to be a deleterious determinant of 1-year mortality after stroke [3].

Our findings have important implications for post-acute stroke care organization. Both secondary prevention and medical follow-up are crucial to reduce recurrence and complications after ischemic stroke. Approximately one-third of patients with stroke are re-hospitalized within the first year following the event, with subsequent increase in mortality [21]. Strategies that aim to reduce readmission should be a major focus of post-stroke care, and specific regular follow-up interventions after discharge could be of relevance in achieving this goal [22]. Looking at our results, deprived patients may appear particularly suited to such interventions, and further studies are needed to explore whether they could benefit from dedicated post-stroke supporting programs. If confirmed, early identification of deprivation will constitute a major challenge in stroke care strategies.

The main strength of our study is the use of an individual score to assess deprivation. The EPICES score explores many more dimensions than only isolated socioeconomic variables, and has been validated with good reliability compared with other aggregated deprivation indexes [17]. Although ethnic and cultural background was contrasted between centers, we have previously shown that the EPICES score could be used to assess precariousness and social deprivation between contrasted areas [23]. The cut-off of 30.17 that we used to define patients as deprived corresponds to the lower boundary of the fourth quintile in the validation study, and is used in most of the studies using this score. Our results were consistent when the EPICES score was considered as a continuous variable in sensitivity analysis. In addition, our cohort excluded <2% of patients because of missing data for the EPICES score. The models retained were based on a compromise between clinical and statistical relevance in order to better understand the EPICES effect on mortality. The impact of deprivation of delayed survival for patients with ischemic stroke was consistently observed, and only marginally modified after further adjustment for study level, pre-stroke mRS, thrombolysis and gender. Conversely, several limitations must be acknowledged. First, we cannot exclude the possibility that residual confounding such as unmeasured comorbidities may account for the observed associations. In addition, the cumulative risk of death 1 month after stroke was lower than the rates observed in several hospital-based stroke studies. It could be assumed that these numbers might be due to the fact that some patients with very severe stroke may not have been included based on the judgment of the investigators about poor early prognosis and inability to evaluate deprivation as stipulated in our study protocol. However, in our study, the median NIHSS score of patients with intracerebral hemorrhage (10 [IQR 3–16], data not shown) was similar to that reported in population-based registries [24], and clinical severity of ischemic stroke patients (median NIHSS score 6 [IQR 3–12]) was even greater than that noted in these registries. Of note, as it could be assumed that the more severe patients at admission were the most deprived [15], this bias may have contributed to an underestimation of the association between deprivation and risk of death. Furthermore, the lower cumulative risk of death could be partially explained by an age effect. Hence, patients included in our study were younger than those enrolled in population-based registries from high-income

countries [25], mainly because of the hospital-based nature of the recruitment, the fact that about 70% of patients were included in French Overseas Departments where mean age at stroke onset is lower than in mainland France [26], and a possible non-inclusion of patients with pre-stroke severe dementia, especially those living in an long-term care institution, in whom social deprivation is difficult to assess, and who have a poorer vital prognosis [27]. Finally, the lower number of events than expected could have led to a lack of power, especially for intracerebral hemorrhage (10% power to detect the observed difference between the two groups in death proportions at 1 year).

To conclude, social deprivation was associated with delayed mortality in patients with ischemic stroke and, although the exact underlying mechanisms are still to be identified, our findings suggest that deprived patients could greatly benefit from optimization of post-stroke care. Future work could focus on identifying levers and barriers to adherence to secondary stroke prevention guidelines in this population.

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CONFLICT OF INTEREST

Yannick Béjot received honoraria or consulting fees from AstraZeneca, BMS, Pfizer, Medtronic, MSD France, Amgen, and Boehringer-Ingelheim. Other authors: none. The authors report no disclosures of conflicts of interest.

DATA AVAILABILITY STATEMENT

Data are available on reasonable request due to privacy/ethical restrictions.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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